

The Three-Layer Wall: Authorship-Bias Controls for Generative Agent-Based Simulations

Pranab Sarkar — Independent Researcher — ORCID: 0009-0009-8683-1481

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Abstract

Agent-based simulations built to explore soft-domain questions — cultural transmission, moral behavior, institutional dynamics — face a structural credibility problem: the author writes both the mechanism and the metric, so the simulation can “discover” precisely what its author assumed. This paper presents a practical discipline for building simulations whose results are findings rather than echoes. The core device is the *three-layer wall*: a hard partition of every quantity in the system into fixed ontology, tunable hypotheses, and measured outcomes, with information permitted to flow in only one direction. Around the wall, four operational controls are developed: placebo arms, observability separation with metamorphic audit tests, random-stream isolation for causal attribution, and narration–physics separation. The discipline is demonstrated in a case study: a cosmological agent-based simulation implementing a layered metaphysics as executable mechanism, including a dual-process decision architecture and an endogenous cultural-transmission loop in which agents generate, distort, and are in turn influenced by their own mythology. The controls caught three real authorship-bias defects during development, each of which would otherwise have been reported as a result. Two findings that survived the full battery are reported: living cultural memory exhibits a measurable fidelity half-life (stories under 60 years old matched their source records at 94%; past 150 years, 46%), and narrative influence on behavior is nonlinear — below a threshold exposure strength, stories with 76% population reach produced no measurable behavioral effect, while at full strength the same channel shifted cooperation by +1.9 points and increased the simulation’s terminal attainment rate by 9%, despite negatively-valenced stories dominating the myth pool. The methods are domain-agnostic and inexpensive, and are offered as a checklist for any generative simulation whose author also writes its grading function.

Keywords: agent-based modeling, validation, authorship bias, metamorphic testing, cultural evolution, simulation methodology, placebo controls, model credibility, computational philosophy

1. Introduction

Every generative simulation is an argument written twice: once as mechanism and once as measurement. When the same author writes both, a circularity threatens that is more insidious than ordinary bugs, because its symptom is success. The simulation produces exactly the phenomenon the author hoped to see; the plots are beautiful; nothing crashes. The result is then reported as if the world inside the machine had testified independently, when in fact the author asked a question of their own assumptions and received their own assumptions in reply.

The problem is old — Epstein’s defense of modeling [1] and the pattern-oriented modeling program of Grimm et al. [2] both circle it — but it is acute for a particular class of simulation that has become easy to build and tempting to trust: *generative soft-domain simulations*, in which agents carry psychological or cultural state, mechanisms encode contestable theory (“hardship refines character,” “stories shape conduct”), and the outputs are narratively compelling. Compelling narrative output is precisely the condition under which an author is least likely to audit their own graders, a Goodhart-type failure [3] in which the simulation optimizes the author’s satisfaction rather than the question’s answer.

This paper reports a discipline developed while building such a simulation deliberately — a cosmological agent-based model that takes the layered metaphysics formalized in earlier structural work on simulation-class ontologies [4] and implements it as executable mechanism: persistent agents (“souls”) that carry state across embodied episodes, a dual-process decision architecture, a coupled vice dynamic, and an endogenous mythology loop in which the population generates, distorts, institutionalizes, and is in turn behaviorally influenced by stories about its own history. A system like this is an authorship-bias worst case: every mechanism encodes a philosophical commitment, every metric tempts the author to grade their own thesis, and the output is *maximally* charming. It is therefore a good stress test for credibility controls: anything that keeps this system honest will keep a traffic model honest.

The contributions are: (i) the three-layer wall, a partition-and-flow rule that makes authorship bias a checkable property rather than a virtue of character; (ii) four operational controls that enforce the wall at the code level — placebo arms, observability separation with metamorphic audit tests, random-stream isolation, and narration–physics separation; (iii) a case study in which these controls caught three real defects that would otherwise have been published as findings; and (iv) two positive findings that survived the full battery, reported with the controls that license them.

2. The Three-Layer Wall

Every quantity in the simulation is assigned to exactly one layer:

Layer F — Fixed ontology. The commitments that define *which universe this is*. In the case study: agents persist across episodes; decisions pass through an impulse–awareness–veto pipeline; an exit criterion exists. Changing Layer F is not tuning; it is changing the subject of study.

Layer H — Hypotheses. Named, numeric, sweepable knobs, each phrased as a question rather than an answer: *does hardship refine or damage?* (a scalar multiplying stress-conditioned growth); *do stories shape conduct?* (a scalar on the narrative-exposure channel, sweepable to zero). A hypothesis knob’s default value is a confession, not a conclusion.

Layer M — Measured outcomes. Everything reported: rates, distributions, survival curves, accuracy scores. Layer M is write-only from the simulation’s perspective.

The wall is the flow rule: **F and H may influence the run; M may never.** No rule, no mechanism, no scheduling decision may read a measured value. The rule sounds trivial and is violated constantly in practice, usually through one of three channels: (a) a *grading channel* — the metric quietly encodes the same assumptions as a mechanism, so agreement is tautological; (b) an *oracle*

channel — a mechanism reads state that no in-world entity could observe, smuggling the analyst’s omniscience into the world; (c) a *narration channel* — labels generated for human legibility (“this agent is a sage”) leak back into behavior. The controls below target each channel.

An important clarification: the wall does not forbid *interventions*. An experimenter (or an interactive user) may reach into the world and perturb Layer F/H state at runtime. What it forbids is the *world’s own physics* conditioning on Layer M. Interventions are honest precisely because they are declared as external hands rather than disguised as endogenous law.

3. Four Operational Controls

3.1 Placebo arms

Any claimed effect of an informational mechanism must be compared against a *shape-matched placebo*, not only against absence. In a companion experiment on memory-augmented language-model generation, a treatment arm injecting twelve retrieved craft rules outperformed a no-injection baseline by 26 percentage points on a compliance rubric — but a placebo arm injecting twelve length-matched, framing-matched platitudes captured 9 of those points. A third of the naive effect was prompt mass, not knowledge. The general rule: when the treatment is *content*, the control must be *content-shaped noise*. Reporting treatment-versus-nothing for informational interventions should be regarded as under-controlled by default.

3.2 Observability separation and metamorphic audit tests

The oracle channel is subtlest where a simulation contains both an inner truth and an outer appearance. In the case study, the decision pipeline distinguishes inwardly *effortless* right action from right action won by costly self-override — but no in-world observer can see that difference; a witness sees only the outward act. When the cultural loop was first designed, story-formation was keyed to the analyst’s event registry, which recorded inner difficulty and cross-episode identity. This was an oracle: the culture would have “remembered” precisely those deeds the metaphysics deemed significant, and the finding “cultural fame tracks inner merit” would have been prewritten.

The repair is a schema separation. A causal **public trace** carries only what witnesses could observe (actor’s public name, situation type, outward act, apparent scale, witness count), with publicness sampled from situational observability, never from inner significance. A noncausal **audit record** carries the inner truth (agent identity across episodes, inner decision type) for analytics only. The invariant is then *metamorphically testable* [5]: perturbing or destroying audit-only fields mid-run must leave the world’s trajectory bit-identical. In the case study this test is run by vandalizing the audit log every simulated year in a paired run and asserting identical population, attainment, and myth histories. The test failed on first execution — not through an oracle, but by exposing a latent nondeterminism (unseeded unique identifiers perturbing set-iteration order), itself a credibility defect, since claimed “paired runs” had not been pairs at all. Both defects were fixed; the test now passes and runs as part of the experiment harness.

3.3 Random-stream isolation

Dose-response comparisons attribute outcome differences to a mechanism's strength. But if all mechanisms draw from one random stream, arms differ not only in the mechanism under test but in *dice consumption*: a channel that merely rolls random numbers shifts every subsequent draw, and chaotic divergence masquerades as effect. The case study therefore assigns the cultural subsystem its own generator, seeded independently of the core life-cycle stream. With streams isolated, a swept channel can be attenuated to zero without perturbing any draw outside itself; behavioral differences between arms then flow only through the mechanism's declared causal path. Stream isolation is cheap, and without it, single-seed intervention studies in coupled agent systems are largely theater.

3.4 Narration–physics separation

Generative simulations produce human-facing narration: role labels, event prose, biography summaries. The temptation is to reuse these labels as state. The rule: narration is computed *from* physics and never read *by* physics, enforced by a renaming invariance — any narration string may be arbitrarily relabeled without changing the trajectory. In the case study, derived life roles (“mystic,” “reformer”) appear in agent biographies and nowhere in any mechanism.

3.5 Adversarial design review

As a process control, the design of each new mechanism was subjected to red-team review by an independent large language model prompted specifically to locate wall violations and oracle channels before implementation. This practice located the observability defect of §3.2 at the design stage. The practice is noted because it is cheap, repeatable, and materially changed the shipped architecture; model-assisted red-teaming of *designs* (as distinct from code) appears underused in the simulation literature.

4. Case Study

The host system implements, as executable mechanism, the layered ontology analyzed structurally in [4]: persistent agents cycle through embodied episodes; an exit criterion (“liberation”) requires not only optimal state but demonstrated robustness across all stress conditions and *effortlessness* — the agent's unmediated first impulses must themselves be aligned, so that suppression is distinguishable from transformation. Decisions pass through a dual-process pipeline: conditioned impulse assembly (weighted by habit traces that deepen with use and decay between episodes), a stochastic awareness gate, and a costly self-override contest whose faculty strengthens only through exercise and erodes under witnessed self-betrayal. A coupled vice dynamic makes moral failure a feedback attractor rather than a low score. None of these commitments is defended here; they are Layer F, and the point is methodological: a maximally tempting system kept honest.

The cultural loop closes the largest cycle: outward acts probabilistically leave public traces; traces are narrated by witnesses (and fabricated by motivated liars — fabrication is an agent act driven by agent state, not a scheduler event); narrations become myths represented as *claim bundles* (actor claim, event claim on a categorical scale ladder, doctrinal propositions) whose components mutate independently through retelling — misattribution, scale drift, moral inversion, collective-to-hero compression, and annexation of doctrine by unrelated stories; myths compete for finite attention,

and heard stories lay weak, episode-local habit traces in the same representational slot as personally acquired habit — increasing an act’s availability, never its alignment. Because the simulation’s physics is the ground truth its own mythology describes, every myth is gradeable: doctrinal accuracy, doctrinal coverage, and historical fidelity are computed against the noncausal audit archive.

Three defects caught by the controls during development, each of which would otherwise have shipped inside a result:

1. **A grading-channel defect** (companion experiment): a compliance rubric written by the same hand that wrote the injected guidance, discovered when an obviously degenerate output scored 11.8% by vacuously passing the rubric’s negative checks. Repair: validity gates before scoring; and for the main claim, replacement of the authored rubric with a spec the generator could not have seen.
2. **An oracle-channel defect** (§3.2): story formation keyed to inner significance. Caught at design review; repaired by observability separation.
3. **A pairing defect**: unseeded identifiers made “identical seeds” non-identical, invalidating paired-run claims silently. Caught by the metamorphic audit test; repaired with deterministic serials and ordered iteration at every random-consuming boundary.

5. Findings That Survived

Two results are reported *because* the battery licenses them; the controls they passed are listed with each.

5.1 Living cultural memory has a fidelity half-life. Grading each living story-myth against its archived source trace: myths younger than 60 simulated years matched their sources at 0.94 (actor, act-valence, scale, plurality); myths older than 150 years matched at 0.46, with fabricated stories (no source event) additionally occupying a large share of the high-popularity tail. Doctrinal coverage remained at 1.0 throughout — every ground-truth proposition remained taught by some living tradition even as the biographies carrying doctrine rotted. Distortion operators were not fitted to produce this; decay-with-age is emergent from uniform per-retelling mutation exposure. *Controls passed*: audit noncausality (metamorphic test), narration separation; the fidelity metric reads the analytics archive only.

5.2 Narrative influence on conduct is a threshold, not a dial. Sweeping the exposure-strength knob across {0, 0.5, 1.0} (three seeds, 350 years, 800 agents, isolated cultural stream): at 0.5, with 76% of the adult population carrying story-laid habit traces, no behavioral metric moved (cooperation $\Delta -0.001$). At 1.0, cooperation rose +1.9 points, violence fell -1.9, effortless action rose +2.0, and terminal attainment rose 9% — despite negatively-valenced stories holding a majority of believers. The sign asymmetry has a mechanistic reading visible in the decision pipeline: negative stories groove acts that unrefined agents’ impulses already dominate (redundant at the argmax), while positive stories can flip marginal agents (pivotal). Stated as a portable hypothesis: *exemplar narratives act on the margin, not the mode; and below a salience threshold, saturating reach produces no behavioral effect*. *Controls passed*: stream isolation (arms differ only in the declared

channel), placebo logic (the 0-arm is a true mechanism ablation, not merely content absence), narration separation.

Neither finding is offered as a claim about human societies. Both are offered as cheaply generated, control-hardened hypotheses of the kind such simulations are for [1]: precise enough to look for in empirical data on oral-tradition fidelity and narrative persuasion, and honest about their provenance.

6. Related Work

Pattern-oriented modeling [2] disciplines model *structure* against observed patterns; the present work is complementary, disciplining the *authorship relation* between mechanism and metric. Classic cultural-transmission models [6, 7] established mutation-and-selection dynamics for traits and opinions; the case study differs in grading transmitted content against an internal ground truth, which is only possible when the modeled world’s physics is owned — a methodological advantage of simulation unavailable to field studies, and one that authorship controls are required to spend honestly. Metamorphic testing [5] supplies the template for the audit invariants. The memory substrate underlying the persistent-agent implementation is described in [8].

7. Limitations

The case-study effects are small-n (three seeds per arm) and single-parameterization; the wall governs *validity*, not statistical power. The placebo principle is demonstrated in a companion language-model experiment rather than in-simulation. Surgical single-myth deletion — proving that one specific false story altered history — remains confounded by chaotic divergence even under stream isolation and is left as future work requiring larger seed batteries. Finally, the wall constrains what a simulation may claim, not what an author may believe; it is a hygiene practice, not an epistemology.

8. Conclusion

A generative simulation is trustworthy in proportion to the difficulty of lying to oneself with it. The three-layer wall and its four controls make several common self-deceptions mechanically checkable: tautological grading (placebo arms), smuggled omniscience (observability separation with metamorphic tests), chaos dressed as effect (stream isolation), and narration leaking into law (renaming invariance). In the case study these controls converted three would-be findings into three found bugs — which is the method working, since a discipline that never catches its author is decoration. What remains after the battery is smaller than what was hoped and more likely to be true, which is the correct direction of trade.

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Code Availability

The complete engine, experiment harnesses, wall tests, and interactive visualizations are open-source at <https://github.com/spranab/soulsim> (v1.0, the version described here).